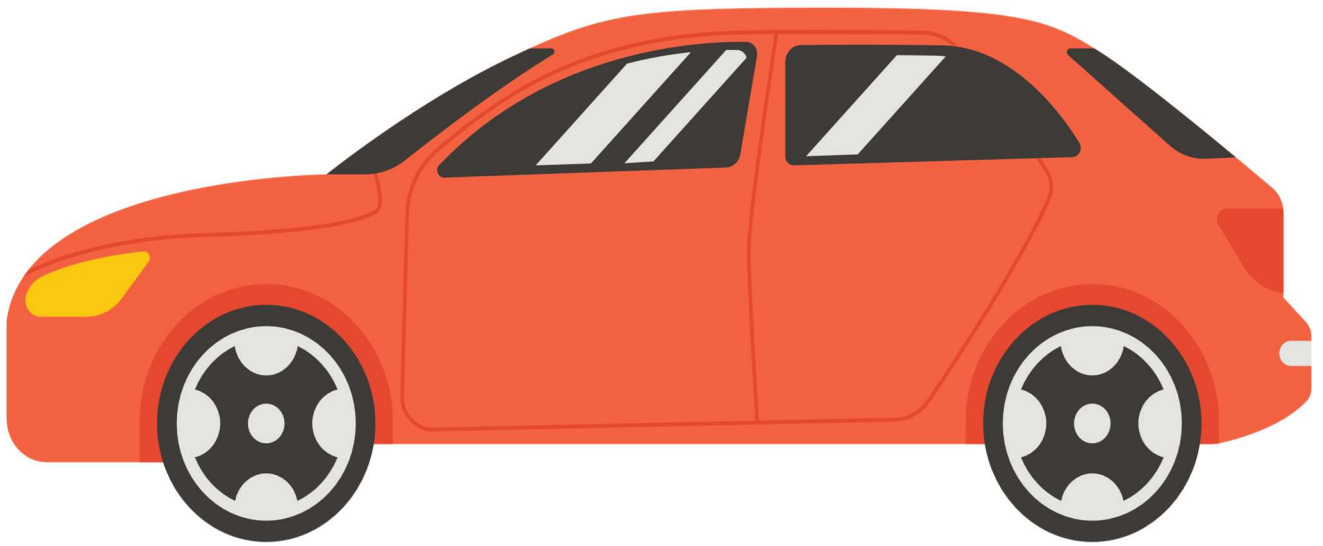


Global Road Accident Report

Comprehensive Insights into Accident Hotspots, Costs, and Causes



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Introduction

Road accidents remain one of the leading causes of injury and death worldwide, affecting millions of people every year. Understanding the patterns, causes, and economic impact of road accidents is crucial for developing effective safety measures. Our analysis, based on a comprehensive dataset of 132,000 global accident cases from 2000 to 2024, offers insights into accident hotspots, contributing factors, and potential solutions.

Describe the Dataset and Source

The dataset (“road_accident_dataset”) contains 132K records of global accidents.

It includes attributes like accident location (country, weather conditions), severity, driver demographics, and economic loss estimates.

We obtained it from an open data portal (or a public repository like Kaggle), chosen for its broad coverage and detail.

Why This Dataset Was Chosen

We wanted to analyze global accident patterns—examining how geography, weather, and driver factors contribute to accidents.

The dataset's rich features (severity, cost, weather) allow for deeper insights and potential applications in road safety policy.

Questions For Global Road Accident

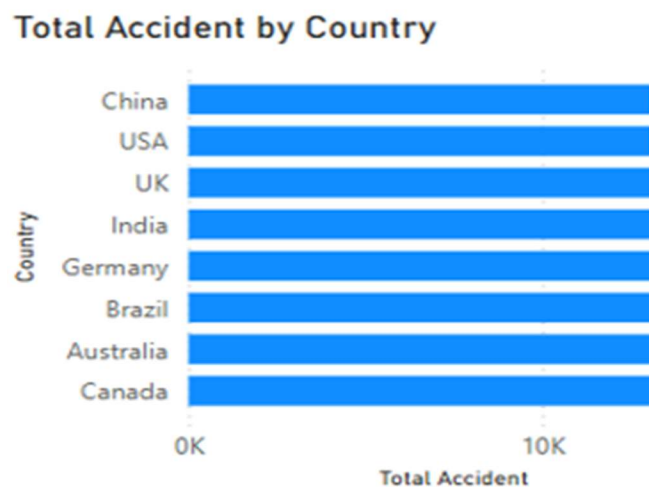
1. Which regions experience the highest number of road accidents?
2. Which weather conditions contribute to the highest number of accidents?
3. What are the trends in fatalities occurrences over time?
4. What is the total economic impact of road accidents, and how significant are the financial losses?
5. Is there a noticeable difference in accident involvement between male and female drivers?

How We Connected to the Data Using Power BI

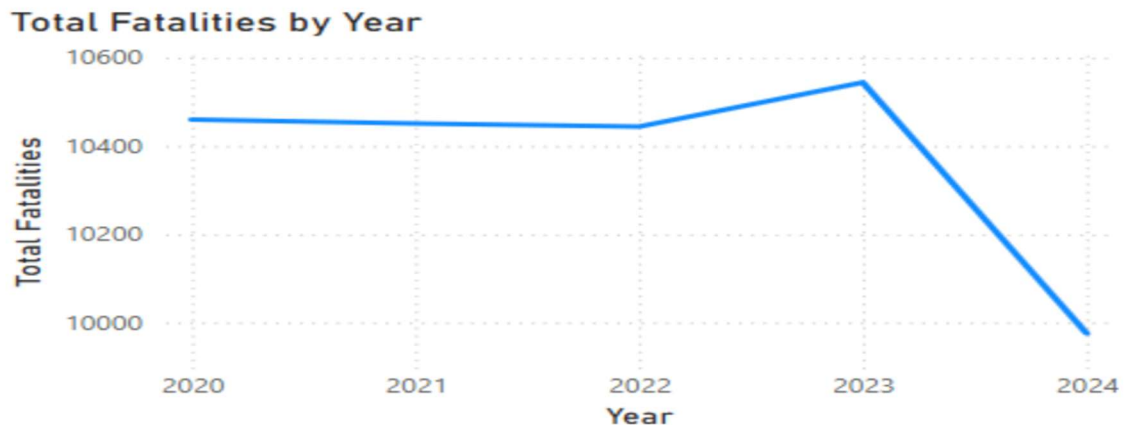
- We imported the CSV file into Power BI Desktop.
Used Power Query to clean/transform data (fix data types).
- Created DAX measures (e.g., Total Accidents, Fatalities, Economic Loss) and built interactive visuals (maps, bar charts, etc.).
- Where Do Most Accidents Occur?

Visuals

Total Accident by Country

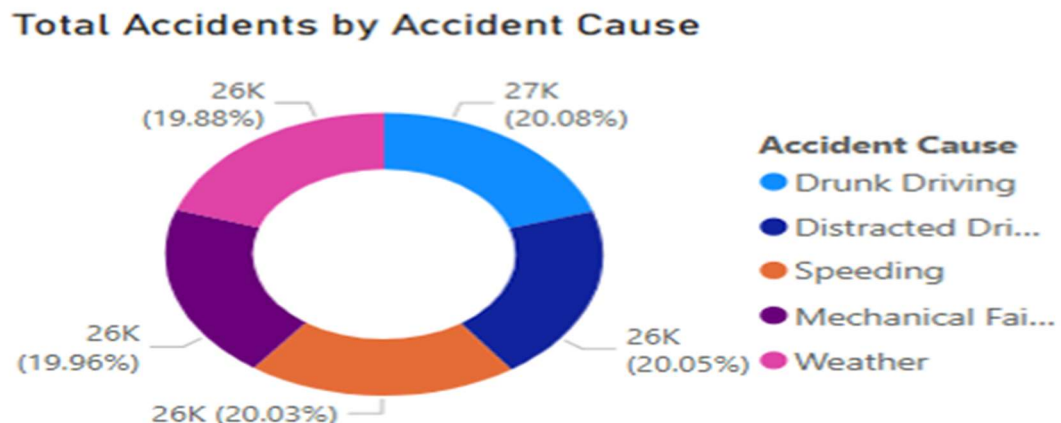


Canada ranks highest in accident counts. Possible reasons (harsh weather, population density, infrastructure). Data Source: Accidents by Country.



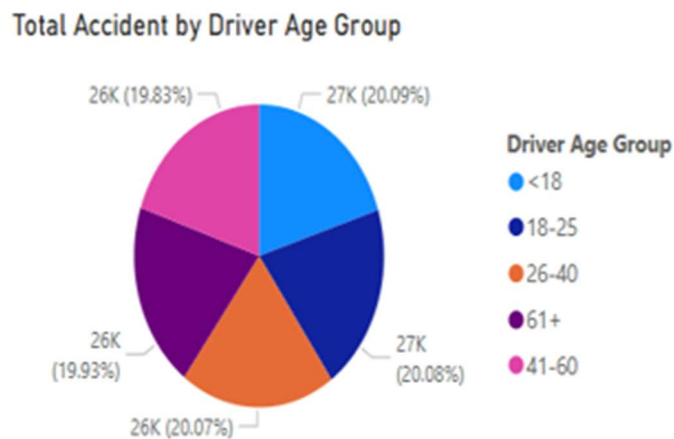
Trends Over Time

The number of total fatalities remained relatively stable between 2020 and 2022 with minor fluctuations. A slight increase is observed in 2023, suggesting a temporary rise in fatal accident occurrences. Significant decline in 2024, showing a sharp drop in fatalities compared to previous years.



Total Accidents by Accident Cause

The chart shows the total number of accidents by cause, with drunk driving, speeding, distracted driving, mechanical failure, and weather-related issues all contributing nearly equally, each causing around 26K to 27K accidents, making them significant factors in road accidents.



Total Accident by Driver Age Group

The chart shows that drivers of all ages have a similar number of accidents, with the 18-25 and 26-40 age groups having slightly more (27K), while other groups, including under 18, 41-60, and 61+, have around 26K accidents each.

Time of Day	Accident no
Afternoon	32960
Evening	33021
Morning	32788
Night	33231

Time Of the Day Accident

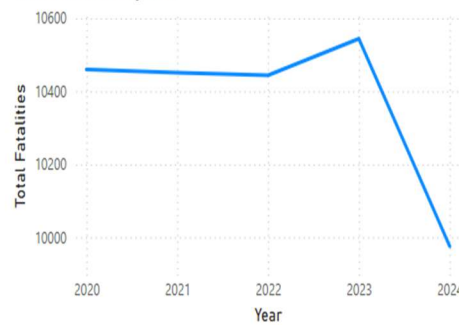
The table shows the number of accidents at different times of the day. Night has the highest accidents (33,231), followed by evening (33,021), afternoon (32,960), and morning (32,788). This means accidents happen at all times, but they are slightly more common at night.

GLOBAL CAR ACCIDENT DASHBOARD

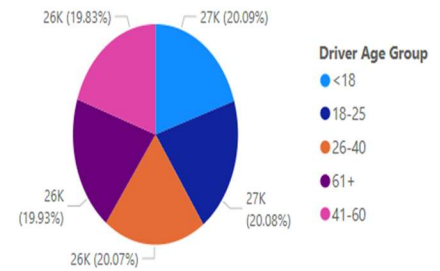
Car Accident



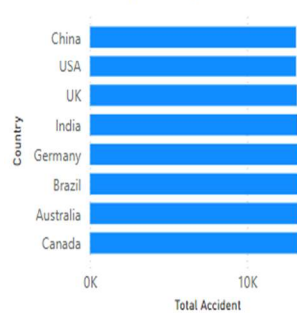
Total Fatalities by Year



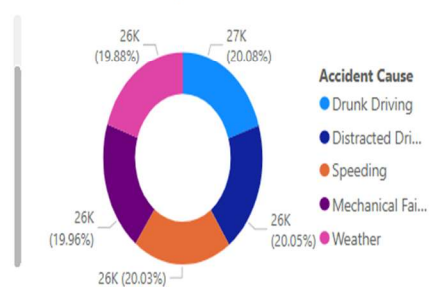
Total Accident by Driver Age Group



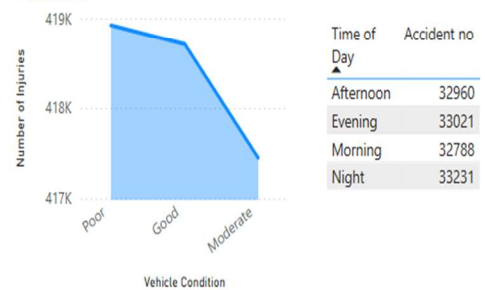
Total Accident by Country



Total Accidents by Accident Cause



Number of Injuries by Vehicle Condition



Conclusion

Overall, the Global Road Accident Report reveals that accidents remain significant and steady over time, incurring high economic losses (7bn) and a notable fatality count (263K).

Windy conditions emerge as a top environmental risk, while both genders share similar accident rates. Regionally, Canada stands out for elevated accident counts.

These insights highlight the need for targeted safety efforts—such as better infrastructure and driver education in adverse weather, stricter enforcement of safe driving practices, and attention to specific causes like mechanical failures—to reduce both human and economic costs globally.